

Makhura Kamogelo

Week 7 practical

Tasks:

Step 1: Standard ACL — Only SALES May Reach the Server

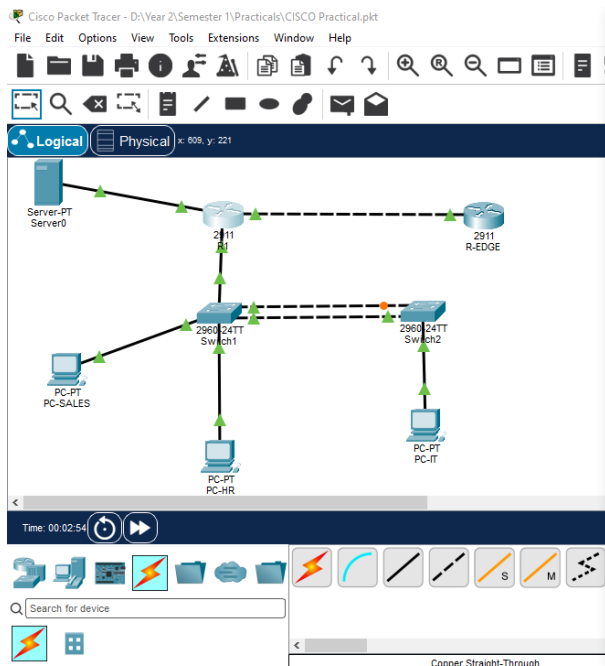
1. access-list 10 permit 192.168.10.0 0.0.0.255 access-list 10 deny any
2. Apply inbound on the interface facing the server: interface g0/1 ip access-group 10 in
3. show access-lists

Step 2: Extended ACL — HR May Only Use HTTP to the Server

1. access-list 100 permit tcp 192.168.20.0 0.0.0.255 host [server IP] eq 80 access-list 100 deny ip any host [server IP] access-list 100 permit ip any any

Step 3: SSH Hardening

1. hostname R1 ip domain-name net621.local crypto key generate rsa username admin secret Cisco123!
2. line vty 0 4 transport input ssh login local
3. Confirm Telnet is now refused, and connect successfully via SSH: ssh -l admin [R1 IP]



R1

Physical Config CLI Attributes

IOS Command Line Interface

```
Cisco IOS2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400K
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
256K bytes of non-volatile configuration memory.
249956K bytes of ATA System CompactFlash 0 (Read/Write)

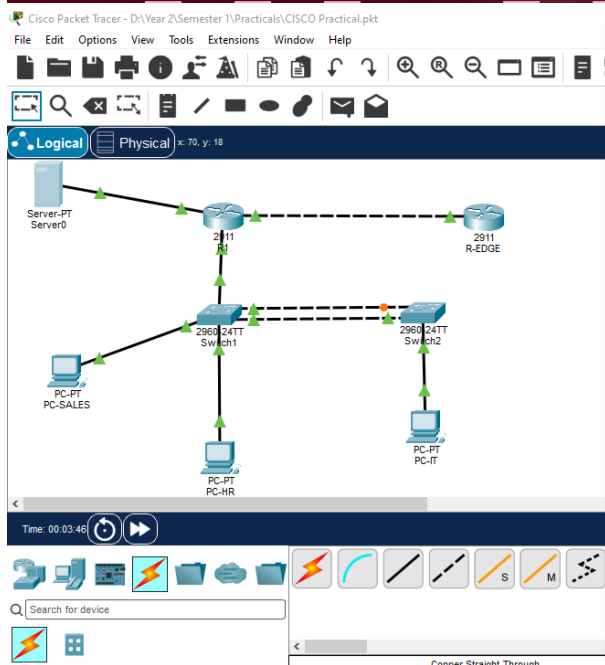
Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/2
Router(config-if)#ip address 192.168.100.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
Router(config-if)#exit
Router(config)#
Router(config)#
```

Copy Paste

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Server0

Physical Config Services Desktop Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.100.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.100.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::201:96FF:FE6D:2167

Default Gateway:

DNS Server:

802.1X

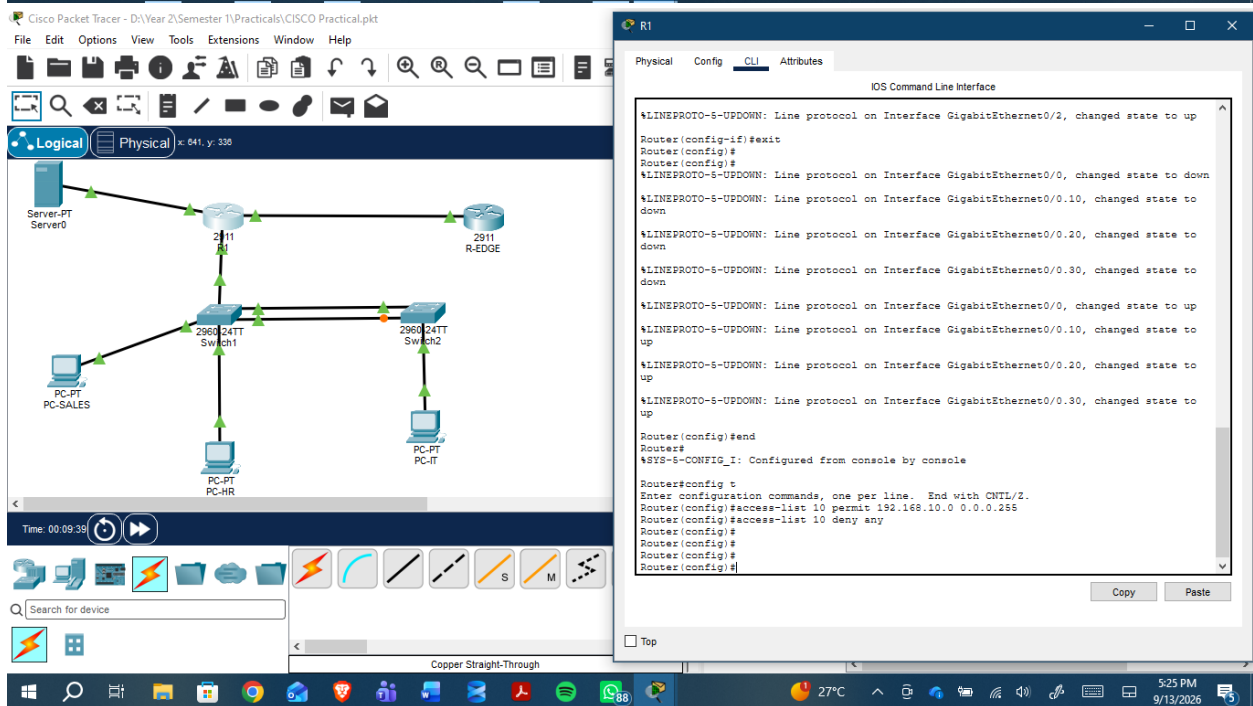
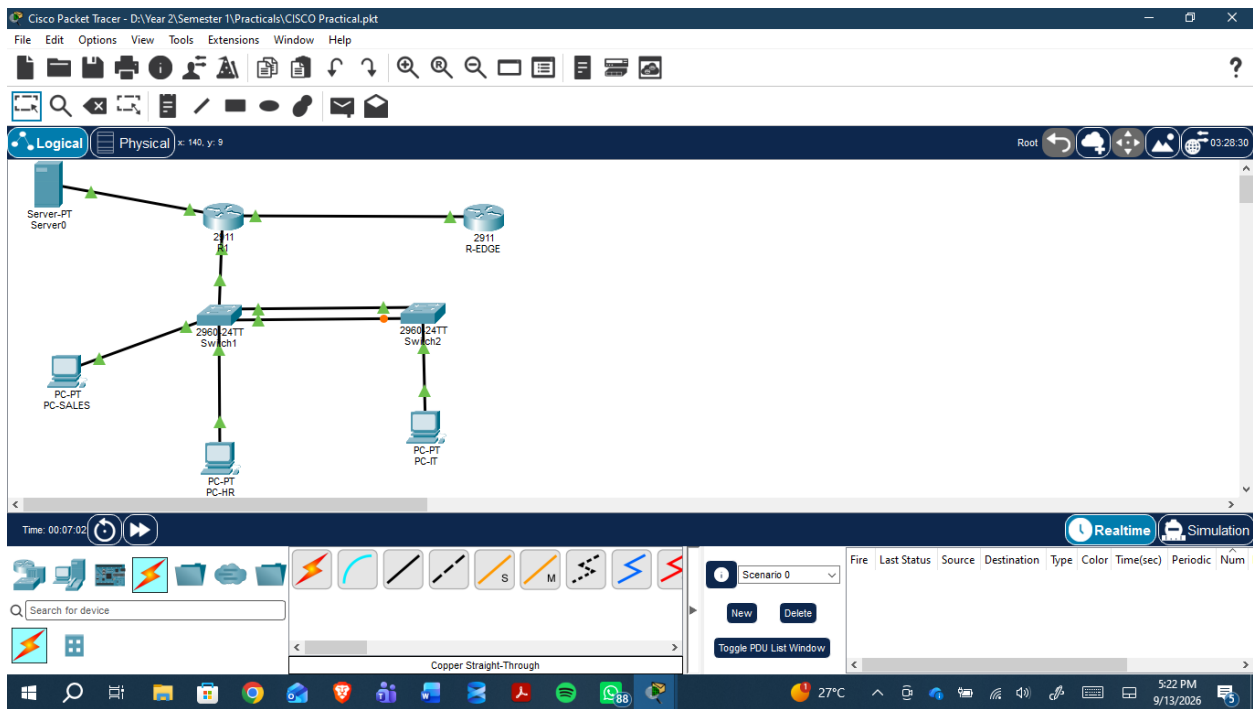
☐ Use 802.1X Security

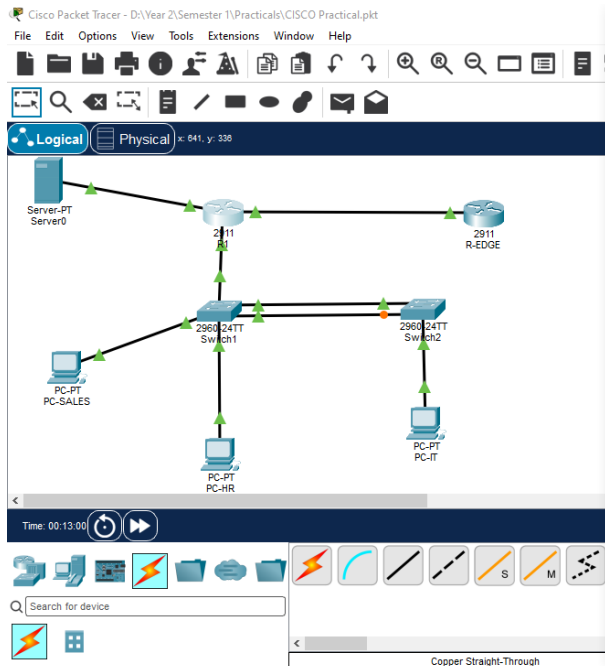
Authentication: MDS

Username:

Password:

Top





R1

Physical Config CLI Attributes

IOS Command Line Interface

```
!LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to down
!LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
!LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
!LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
!LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

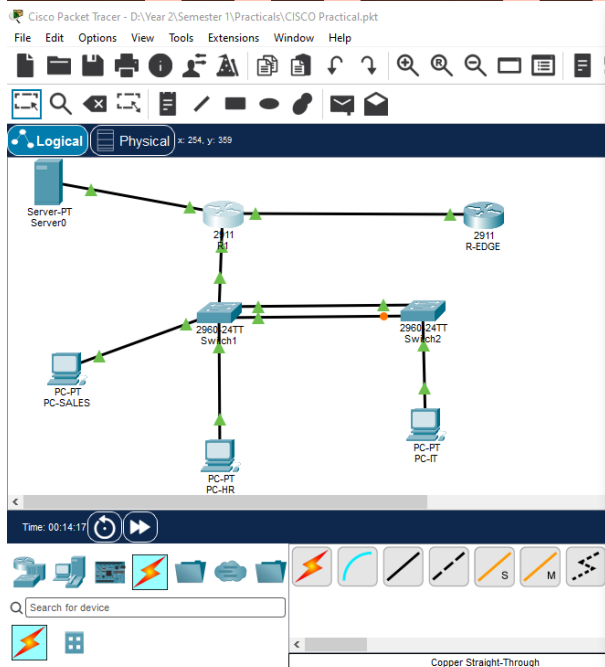
Router(config)#end
Router#
!SYS-5-CONFIG_I: Configured from console by console

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 10 permit 192.168.10.0 0.0.0.255
Router(config)#
Router(config)#
Router(config)#
Router(config)#exit
Router#
!SYS-5-CONFIG_I: Configured from console by console

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#exit
Router(config)#interface g0/2
Router(config-if)#description LINK TO SERVER
Router(config-if)#ip access-group 10 in
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#
```

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R1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#
Router#show access-lists
Standard IP access list 10
 10 permit 192.168.10.0 0.0.0.255
 20 deny any

Router#show ip interface g0/2
GigabitEthernet0/2 is up, line protocol is up (connected)
Internet address is 192.168.100.1/24
Broadcast address is 255.255.255.255
Address determined by setup command
MTU is 1500 bytes
Helper address is not set
Directed broadcast forwarding is disabled
Outgoing access list is not set
Inbound access list is 10
Proxy ARP is enabled
Security level is default
Split horizon is enabled
ICMP redirects are always sent
ICMP unreachable are always sent
ICMP mask replies are never sent
IP fast switching is disabled
IP fast switching on the same interface is disabled
IP Flow switching is disabled
IP Fast switching turbo vector
IP multicast fast switching is disabled
IP multicast distributed fast switching is disabled
Router Discovery is disabled
IP output packet accounting is disabled
IP access violation accounting is disabled
TCP/IP header compression is disabled
RTP/IP header compression is disabled
Probe proxy name replies are disabled
Policy routing is disabled
Network address translation is disabled
BGP Policy Mapping is disabled
Input features: NCL Check
WCCP Redirect outbound is disabled
WCCP Redirect inbound is disabled
```

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Cisco Packet Tracer - D:\Year 2\Semester 1\Practicals\CISCO Practical.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 254, y: 359

Time: 00:14:23

Search for device

Copper Straight-Through

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#show ip interface g0/2
GigabitEthernet0/2 is up, line protocol is up (connected)
Internet address is 192.168.100.1/24
Broadcast address is 255.255.255.255
Address determined by setup command
MTU is 1500 bytes
Helper address is not set
Directed broadcast forwarding is disabled
Outgoing access list is not set
Inbound access list is 10
Proxy ARP is enabled
Security level is default
Split horizon is enabled
ICMP redirects are always sent
ICMP unreachable are always sent
ICMP mask replies are never sent
IP fast switching is disabled
IP fast switching on the same interface is disabled
IP Flow switching is disabled
IP Fast switching turbo vector
IP multicast fast switching is disabled
IP multicast distributed fast switching is disabled
Router Discovery is disabled
IP output packet accounting is disabled
IP access violation accounting is disabled
TCP/IP header compression is disabled
RTP/IP header compression is disabled
Probe proxy name replies are disabled
Policy routing is disabled
Network address translation is disabled
BGP Policy Mapping is disabled
Input Features: MCL Check
WCCP Redirect outbound is disabled
WCCP Redirect inbound is disabled
WCCP Redirect exclude is disabled

Router#
Router#
```

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Cisco Packet Tracer - D:\Year 2\Semester 1\Practicals\CISCO Practical.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 583, y: 183

Time: 00:23:28

Search for device

Copper Straight-Through

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
WCCP Redirect exclude is disabled

Router#
Router#
Router#
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#no access-list 10
Router(config)#interface g0/1
Router(config-if)#no ip access-group 10 in
Router(config-if)#exit
Router(config)#access-list 100 permit top 192.168.20.0 0.0.0.255 host 192.168.100.10 eq 80
Router(config)#access-list 100 deny ip any host 192.168.100.10
Router(config)#access-list 100 permit ip any any
Router(config)#
Router(config)#
Router(config)#access-list 100 permit ip 192.168.10.0 0.0.0.255 host 192.168.100.10
Router(config)#access-list 100 permit top 192.168.20.0 0.0.0.255 host 192.168.100.10 eq 80
Router(config)#access-list 100 deny ip any host 192.168.100.10
Router(config)#
Router(config)#
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#access-list 100 permit ip any any
Router(config-if)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

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Cisco Packet Tracer - D:\Year 2\Semester 1\Practicals\CISCO Practical.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 593, y: 381

Server-PT Server0 2911 R-EDGE 2960 24TT Switch1 2960 24TT Switch2 PC-PT PC-SALES PC-PT PC-IT PC-PT PC-HR

Time: 00:23:53

Search for device

Copper Straight-Through

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#interface g0/1
Router(config-if)#no ip access-group 10 in
Router(config-if)#exit
Router(config)#access-list 100 permit tcp 192.168.20.0 0.0.0.255 host 192.168.100.10 eq 80
Router(config)#access-list 100 deny ip any host 192.168.100.10
Router(config)#access-list 100 permit ip any any
Router(config)#
Router(config)#access-list 100 permit ip 192.168.10.0 0.0.0.255 host 192.168.100.10
Router(config)#access-list 100 permit tcp 192.168.20.0 0.0.0.255 host 192.168.100.10 eq 80
Router(config)#access-list 100 deny ip any host 192.168.100.10
Router(config)#access-list 100 permit ip any any
Router(config)#
Router(config)#exit
Router#
$SYS-5-CONFIG_I: Configured from console by console
Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#access-list 100 permit ip any any
Router(config-if)#exit
Router#
$SYS-5-CONFIG_I: Configured from console by console
Router#
Router#
Router#show access-lists 100
Extended IP access list 100
    permit tcp 192.168.20.0 0.0.0.255 host 192.168.100.10 eq www
    deny ip any host 192.168.100.10
    permit ip any any
    permit ip 192.168.10.0 0.0.0.255 host 192.168.100.10
Router#
Router#
```

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File Edit Options View Tools Extensions Window Help

Logical Physical x: 589, y: 349

Server-PT Server0 2911 R-EDGE 2960 24TT Switch1 2960 24TT Switch2 PC-PT PC-SALES PC-PT PC-IT PC-PT PC-HR

Time: 00:30:29

Search for device

Copper Straight-Through

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/2
Router(config-if)# no ip access-group 10 in
Router(config-if)# no ip access-group 10 out
Router(config-if)# exit
Router(config)# no access-list 10
Router(config)# end
Router#
$SYS-5-CONFIG_I: Configured from console by console
Router#
Router#
Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#ip domain-name net21.local
R1(config)#crypto key generate rsa
The name for the keys will be: R1.net21.local
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
How many bits in the modulus [512]: 1 When asked How many bits: 1024
# A decimal number between 360 and 4096
How many bits in the modulus [512]: # A decimal number between 360 and 4096
How many bits in the modulus [512]:
# Generating 512 bit RSA keys, keys will be non-exportable...[OK]
R1(config)#username admin secret Cisc0ol23!
*Mar 1 4:43:10: RSA Key size needs to be at least 768 bits for ssh version 2
*Mar 1 4:43:10: SSH-5-ENABLED: SSH 1.5 has been enabled
R1(config)#
R1(config)#line vty 0 4
R1(config-line)# transport input ssh
R1(config-line)# login local
R1(config-line)# exit
R1(config)#
```

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Reflection Questions (answer these in your submission):

1. Why is an extended ACL more suitable than a standard ACL for the HR-to-server rule?

An extended ACL (Access Control List) is more suitable than a standard ACL for the HR-to-server rule because it gives you more specific control over network traffic.

An extended ACL is more suitable because it provides granular control over traffic. It can specify both the source and destination IP addresses, as well as the protocol and port number. This allows the network administrator to permit HR users to access only the required server services while blocking unauthorized traffic. A standard ACL only filters based on the source IP address, making it less precise.